🚀 AI Career Roadmap

Personalized Strategy Report — Student → Data Scientist / AI Engineer

3-MONTH SPRINT PLAN

🚀 Current Position

Student with working knowledge of **Python & Data Analysis**, plus early exposure to **AI Engineering** concepts (LLMs/APIs). No formal industry project portfolio yet.

🎯 Target Goal

Become a **Data Scientist / AI Engineer** within **3 months** — build a job-ready portfolio, core ML/AI skills, and an active professional network.

📈 Skill Gap Analysis

Category	You Have	Gap to Close
Programming	Python, Data Analysis	OOP for production code, SQL fluency
Core ML	Basic stats/pandas	Scikit-learn, Model evaluation, Feature engineering
AI Engineering	LLM/API exposure	RAG pipelines, LangChain/vector DBs, Prompt eval
Deployment	—	Git/GitHub, APIs (FastAPI), Cloud basics
Portfolio & Brand	—	3 showcase projects, LinkedIn presence, Resume/GitHub polish

🛠 Recommended Learning Plan (Priority Order)

Phase	Focus	Key Topics
Weeks 1–2	Foundations Refresh	Advanced Python (OOP), SQL, Statistics for ML, Git/GitHub workflow
Weeks 3–5	Core Machine Learning	Scikit-learn, regression/classification, model evaluation, feature engineering, Kaggle practice
Weeks 6–8	AI Engineering	LLM APIs (OpenAI/Anthropic), LangChain, RAG pipelines, vector databases (Pinecone/Chroma)
Weeks 9–10	Deployment & MLOps Basics	FastAPI, Docker basics, Streamlit apps, cloud deployment (AWS/GCP free tier)
Weeks 11–12	Portfolio & Job Prep	Polish GitHub/resume, mock interviews, DSA/ML interview questions, apply + network

💼 Suggested Projects

- **EDA + ML model:** End-to-end prediction project (e.g., pricing/churn) with clean GitHub repo & README
- **RAG chatbot:** Document Q&A app using LangChain + vector DB, deployed via Streamlit
- **AI Agent/API tool:** Small FastAPI service wrapping an LLM for a real use case
- **Capstone:** Combine ML + LLM (e.g., AI-powered data analyst assistant) — flagship portfolio piece

🌐 Networking Strategy

- Post weekly on LinkedIn: build-in-public updates on projects/learning
- Engage with 5–10 AI/DS professionals' posts daily; join 2–3 niche communities (Discord/Slack)
- Attend 1 virtual meetup/webinar per month; connect with speakers/attendees
- Reach out to 5 professionals/week for informational chats; ask for referrals near Month 3

📅 Monthly Milestones

Month 1 — Foundation

- Python/SQL/stats refreshed
- GitHub profile set up
- 1 EDA project shipped

Month 2 — Build

- Core ML mastered
- RAG chatbot deployed
- 50+ LinkedIn connections

Month 3 — Launch

- Capstone project complete
- Resume + portfolio finalized
- Actively applying + interviewing

⚡ Immediate Next Actions (This Week)

1. Set up GitHub + LinkedIn profile with a clear "AI/Data" headline
2. Enroll in one structured ML course (e.g., scikit-learn track) and start daily practice
3. Pick your first EDA dataset and start Project 1
4. Follow 10 AI/DS professionals on LinkedIn and post your Day 1 learning update

Generated using Chain-of-Thought Reasoning